

Fertility and housing: overnight memo

Fit, progress and the economic misses

Best completed primary-weight point: loss **42.282**. This is a provisional searched point, not a certified final calibration.

The no-floor utility generates a large first-birth housing response (1.610 rooms versus 1.465). Joint fit remains incomplete: ownership, wealth and children born by age 25 are too low. Final points reproduce exactly, but the search does not establish an optimum.

Progress: success: 452. Requested capacity: 24 cluster, 10 local. Completed 436 search cases and 16 checks below, plus two cold-start checks. Up to 10 local workers; zero cluster search workers because SSH access expired. All numerical work finished by 07:02 EDT. No model, grid, target or gate changed during search.

Weight system	Search / checks	Loss under main weights
Early fertility weight 3,000	40 / 4	57.608
Unit weights	60 / 4	361.218
Main weights	336 / 8	42.282

Checks include acceptance and final-repeat evaluations. Each row scores the point chosen by that weight system using the main weights. Full re-ranking is saved separately. The target table below reports the main-weight winner.

All 14 restrictions: 13 scored moments and one normalization

Moment	Target	Model	Gap	Weight	Loss
Completed fertility (normalization)	2.1	2.1	1.05e-04	--	--
Childlessness, ages 40-44	0.198	0.202	0.004	3.55e+04	0.612
Exactly one child, among mothers	0.214	0.211	-0.003	2.70e+04	0.237
Mean age at first birth	25.976	26.1	0.123	139.828	2.132
First births at age 30 or older	0.249	0.234	-0.015	1.39e+04	3.177
Aggregate wealth / annual earnings	6.927	6.025	-0.902	7.595	6.178
Annual bequest flow / wealth	0.007	0.007	-1.52e-04	5.17e+06	0.119
Older wealth/income: p90 / median	3.516	3.203	-0.313	10.616	1.042
Mean occupied rooms (AHS)	5.729	5.934	0.205	128.021	5.356
Ownership, ages 30-55	0.676	0.617	-0.059	2339.362	8.114
Housing response to first birth	1.465	1.61	0.145	137.565	2.89
Rooms: 3+ vs 1-2 children at home	0.385	0.296	-0.089	280.528	2.247
Recent-parent ownership gap	0.128	0.119	-0.009	2.71e+04	2.225
Children ever born at 25, capped at 3	0.81	0.528	-0.282	100	7.952

Gap = model - target. Weight and loss use the primary objective. Normalization is unscored. Full-precision source tables are saved beside this memo.

Largest weighted misses: Ownership, ages 30-55: below target by 0.059 (19.2% of loss); Children ever born at 25, capped at 3: below target by 0.282 (18.8% of loss); Aggregate wealth / annual earnings: below target by 0.902 (14.6% of loss).

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Parameters, reliability and the next decisions

Ten fitted parameters: nine searched jointly and one benefit level solved internally to match completed fertility at 2.1. Four additional restrictions do not by themselves establish identification.

Parameter	Estimate	Lower	Upper	Bound / restriction
Housing supply level	6.206	0.2	80	interior
Annual discount factor	0.96	0.94	0.99	interior
Owner housing-service premium	1.089	0.1	5	interior
First-birth fixed cost	0.66	0	8	interior
First-birth taste scale	0.195	0.02	50	near bound
Later-birth taste scale	0.347	0.02	50	near bound
Bequest strength	0.114	0	8	interior
First-child housing loading	0.142	0	0.25	interior
Child-benefit curvature	0.071	0	0.8	interior
Child-benefit level (normalized)	0.144	0	--	positive; normalized

Near bound means within 1% of the permitted interval. The benefit level has a positivity restriction and is fixed by the fertility equation; it has no separately adopted upper search bound.

Reliability, speed and errors

Median accumulated stationary-solve time per completed objective: 6.775 minutes. Selected point has 17/17 standard diagnostic PNGs. All four search runs passed two exact final repeats of all 14 target rows and 31 parameter rows. Final checkpoints and receipts were authenticated; all 17 selected diagnostic plots were inspected. Two extra cold-start checks are outside the table. At the same starting point, warm prices used 13 price evaluations versus 19 cold: 231 versus 312 seconds locally, with moment differences below 1e-5. Legacy Torch replay and cross-host checks remain uncollected.

0 recorded error/exclusion entries; 0 controller heartbeats older than 30 minutes.

What remains uncertain

- Bequests count all positive estates against a child-directed SCF target. Older-wealth income and first-birth housing-response measurements also retain approximations.
- Weights and the curvature interval remain provisional. The higher early-fertility weight gives 0.544 versus target 0.810, but ownership falls to 0.579; it is not an adopted improvement.
- Preparation failures: one zero-solve cluster submission failed; two local smokes were cancelled after source-pin drift. Fresh pinned snapshots replaced them. No accepted result was relabeled.

Next steps

1. Check the age-25 fertility measurement and profile child-benefit curvature with the two fertility taste scales, keeping every fertility target.
2. Review credit, the 2% interest rate, tenure scale, rental options and income/wealth inputs; test the finer housing grid and conception schedule separately.
3. Restore cluster access and collect the legacy/cross-host checks; then run wider searches from several verified starting points under an agreed weighting rule.

Audit trail: summary.json, monitor.json and full-precision CSVs accompany this memo. The supplemental weighted-gap SVG is separate from the unchanged 17 standard diagnostics. This report checks saved tables and receipts; it does not independently re-solve or re-certify checkpoint arrays.